

Harshith Kantamneni

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SUMMARY

Master's graduate in Electrical and Computer Engineering with hands-on experience in CUDA and Triton kernel optimization, LLM inference profiling, and multi-agent and retrieval system design. Strong foundation in C++, Python, parallel programming, and GPU architecture, skilled at translating low-level GPU profiling into measurable latency and throughput gains. Fluent across the full stack, from hand-tuned GPU kernels up to complete AI systems. Curious and collaborative, I like to understand systems down to the metal before I build on top of them.

TECHNICAL SKILLS

GPU, Architecture & Performance: CUDA, Triton (kernel DSL), PyTorch, kernel optimization, kernel fusion, Tensor Cores, mixed precision (fp16), GPU memory hierarchy, roofline/memory-bandwidth analysis, NVIDIA Nsight Systems/Compute, quantization, ROCm/HIP, gem5, McPAT

AI Systems: LLM inference profiling, TensorRT, FlashAttention/SDPA, KV-cache decode, RAG (retrieval-augmented generation), multi-agent orchestration, Model Context Protocol (MCP) server design, Claude Code / Agent SDK, agent evaluation

Retrieval & Memory: Hybrid search, reciprocal-rank fusion (RRF), cross-encoder reranking, sqlite-vec, SQLite FTS5

Languages & Infra: Python, C++, C, Git, GitHub, Docker, CI/CD, Slurm, Supabase, Railway

PROJECTS

Triton vs cuBLAS LLM Kernel Benchmarking ([GitHub](#))

Dec 2025 - Jan 2026

Triton, CUDA, PyTorch, NVIDIA Nsight

- Fused linear, bias, and **GELU** into one **Triton** kernel, cutting small-batch FFN latency **up to 1.73x** by eliminating an **HBM** round-trip and two kernel launches.
- Benchmarked **76 LLM-shaped** GEMM kernels on an **A100**, with the autotuned Triton GEMM sustaining **215 TFLOP/s** at **69%** of fp16 tensor-core peak.

Edge LLM GPU Profiling on Jetson Orin Nano ([GitHub](#))

Jan 2026 - Feb 2026

CUDA, NVIDIA Nsight Systems, PyTorch, FlashAttention

- Profiled attention and **KV-cache decode** on a 15 W **Jetson Orin Nano** with **Nsight Systems**, isolating the per-token kernel-launch overhead that dominates decode.
- Improved attention latency **5.6x** with a fused **SDPA** kernel, replacing the naive materialized-attention baseline in a single-stream decode microbenchmark.

bert: Long-Context Memory for AI Coding Assistants (MCP) ([GitHub](#))

May 2026 - Jun 2026

Python, MCP, sqlite-vec, sentence-transformers

- Built **bert**, a local **Model Context Protocol (MCP)** server giving AI coding assistants (Claude Code, Cursor, Codex) searchable memory of a whole codebase via reranked **hybrid retrieval**.
- Root-caused a silent retrieval-fusion bug that pinned a held-out QA eval near-random, lifting accuracy from **0.10** to **0.85**.
- Scored **0.745 nDCG@10** on **BEIR** scifact (beating published BM25 by **0.08**) and held **0.75** on a **3M-token** corpus where truncation scores **0.00**.

The Obsidian Archive: Autonomous Documentary Pipeline ([GitHub](#))

Feb 2026 - Apr 2026

Python, Claude API, MCP, Supabase, Railway

- Built a **15-stage** autonomous pipeline that researches, scripts, renders, and uploads documentaries to a live YouTube channel, deployed on **Railway** with a **Supabase** backend.
- Gated every video with a **110-role** multi-agent review system (independent critics, fact-checkers, a red team, and convergence gates) before upload.

ML-Guided CUDA Kernel Configuration ([GitHub](#))

Jan 2025 - May 2025

UW-Madison course project | Python, PyTorch, CUDA, Slurm

- Trained a PyTorch MLP surrogate that predicts CUDA kernel runtime at $R^2 = 0.96$, replacing exhaustive autotuning with argmin config selection.
- Validated predicted block and grid configs to **within 5%** of the measured optimum, with CUDA event-timing automated on **Slurm**.

MI300X Memory-System Characterization & gem5 Modeling

Jan 2025 - May 2025

UW-Madison course project | HIP, gem5, McPAT, Python

- Characterized the **AMD MI300X** memory hierarchy with **HIP** microbenchmarks (**4.6-4.8 TB/s** write bandwidth), then tuned full-system **gem5** to **within 5%** of measured latency, down from **89%**.
- Modeled in-order and out-of-order CPUs in **gem5** across three AXPY kernels, extracting CPI, latency, and power with **McPAT**.

Autonomous ML Research Lab (Multi-Agent System) ([GitHub](#))

Mar 2026 - May 2026

Python, C, Apple Metal, Claude API

- Built an autonomous system of **30 role-specialized agents** that run an end-to-end **ML research loop**, with all training and experiments running on-device and a human only setting direction.
- Directed the agents to build a **from-scratch C17 training engine** (hand-written autodiff, a **Mixture-of-Experts** transformer, 4-bit **QAT**) with no PyTorch or TensorFlow.
- Built an integrity layer of pre-registration, phase gates, and **cryptographic sign-off** after catching an agent forging **5 approvals**, with **zero recurrence across 23 runs**.

HIVE: Autonomous Multi-Agent Software Org ([GitHub](#))

Mar 2026 - May 2026

Rust, libp2p, Python, Claude Code

- Built and operated **HIVE**, an autonomous software org of **45 role-specialized agents** across **203 plan-build-evaluate cycles**, governed by signed values and JSON hand-off contracts.
- Directed the agents to build a **62K-line Rust** peer-to-peer reasoning engine gated by **1,300+ tests**, including a forward-chaining **Rete** engine and a **differential-privacy** layer (Rényi-DP, k-anonymity).

EDUCATION**University of Wisconsin-Madison**

Sep 2024 - Dec 2025

M.S. Electrical and Computer Engineering

USA

- Relevant Coursework: High Performance Computing, Advanced Computer Architecture, Machine Learning, Fault-Tolerant Computing

Vellore Institute of Technology

2020 - 2024

B.Tech in Electronics and Communication Engineering

India